Roshan Raj

Curriculum Vitae

Upto May 2024

Personal Details

Name:	Roshan Raj
Nationality:	Indian
City:	Patna, Bihar-800010
Webpage:	kaalkrit.github.io
ORCID iD:	0000-0002-4902-8747
Email iD:	roshankaalkrit@gmail.com, i19ph002@phy.svnit.ac.in

Research interests

Quantum physics, Quantum Field Theory, Relativistic Theories, Reconstruction and Foundations of Quantum Theory, Quantum to Classical Transition, Gravity and EM Unification, Mathematical aspects of Physical Theory (incl. CM, QM and QFT), Mathematical Physics (See: List of problems I resonate with! [Get the PDF])

Educational Details

2019-24	5 years Integrated M.Sc. (Physics), Sardar Vallabhbhai National Institute of Technology, Surat, India. CGPA: **9.33/10** (IX sem)
2016-18	Senior Secondary High Schooling, Loyola High School, CBSE, Patna, India, **77.2%** [Physics, Chemistry, Mathematics, Informatics Practices, English]
2010-16	High Schooling, St. Paul's High School, ICSE, Patna, India, **91.6%**, [Science, Mathematics, Computer App., Social Sci., Hindi, English]

Research Experiences

Master's Dissertation

- Quantum and Classical Applications of Relativistic Relationship of Kinetic Energy and Momentum
  Supervisor: Dr. Vikash K Ojha, SVNIT-Surat; [Jul, 2023-May, 2024]–URL: shorturl.at/1Nug7
  Keywords: Reconstruction of Relativistic Quantum Theory, Relativistic Classical Physics.
  Brief: A newly found classical relation between relativistic kinetic energy K and relativistic momentum $\mathbf{p}$, namely $|\mathbf{p}|^2 = (1+\gamma)mK$, involving Lorentz factor γ, leads to a compact relativistic quantum wave equation $\mathfrak{p}^\mu \partial_\mu \Psi = 0$, Where $\mathfrak{p}^\mu = [(1+\gamma)p_0, i\hbar\gamma\partial_j]$ with Minkowski signature as $(+,-,-,-)$ is modified momentum four-vector. The reported relation is replacing both the Newtonian non-relativistic energy-momentum formula and relativistic total energy-momentum relation, $E^2 = p^2c^2 + m^2c^4$, and that has sought application in both quantum and classical relativistic mechanics. Its application predicted the contractions of Bohr radius at relativistic speeds, corrected the expression of the Relativistic Lagrangian of a free particle, etc.

Institute Mini-projects (e-prints)

3 New light on the concepts of Observable and indeterminacy in the quantum realm
Guide: Dr. Vikash K Ojha, SVNIT-Surat– UG Project III, [Jul-Dec, 2022]–URL: shorturl.at/6lMIS
Keywords: Quantum to Classical Transition
Brief: I argue that observable parameters are crucial for any physical theory, and in quantum mechanics, the imaginary unit 'i' plays a vital role in transforming unobservables into observable complex quantities. In addition, we can have a nondeterministic classical form of the Born-Jordan quantum condition and, thus, a novel reinterpretation of the uncertainty principle.

2 On the investigation of two non-neutral static bodies
Guide: Dr. Sharad K Yadav, SVNIT-Surat– UG Project II, [Jan-May, 2022]– DOI: arXiv.2209.10641

Keywords: Gravity and EM Unification
Brief: This work studies the statics of two charged bodies. It investigates net field strength, defines a function Y resembling a nuclear mass function, and suggests nature's preference for heavier objects at the positive charge centre. It also finds an expression for ellipse geometry and introduces 'quintessence' analogous to electric potential. Finally, it explores the constancy of a factor χ (chi).

1 Dynamics of Two Objects Considering the Minimum Total Potential Energy Principle
Guide: Dr. Himanshu Pandey, SVNIT-Surat– UG Project I, [Jul-Dec, 2021]–DOI: 10.31219/osf.io/ukx9t
Keywords: Gravity and EM Unification
Brief: We proposed a process, namely, "absolute union", that explores two bodies' motion based on minimizing energy. Analyzing this ideal process suggests the existence of electric charges and gravity's alternative role. A mass ratio of two bodies is found to be required to start the process of absolute union, and the law of inertia is explained.

Independent Works (e-prints)

5 The Circle: 7 different perspectives. Magazine (AMaThing 5.0). 2024. URL: shorturl.at/LE30p
Brief: An exploration of the mathematical concept of a circle through various lenses of mathematical disciplines (geometry, algebraic topology, complex numbers, knot theory, etc.)

4 Obtaining the Klein-Gordon Wave Equation Without Using the Quantum Operators. OSF Preprints. 2023. DOI:10.31219/osf.io/6hmyf.
Keywords: Reconstruction and Foundations of Quantum Theory
Brief: We claim that Klein-Gordan (KG) equation should be termed as "Wave equation" and Schrödinger equation (also, Dirac's equation) is proposed to be called as "Governing law". The basis of categorisation is discussed. The obtained expression for massless particles is similar to the Maxwell wave equation.

3 Reviewing Observables in Classical and Quantum Mechanics. OSF Preprints. 2023. DOI:10.31219/osf.io/p2ufx.
Keywords: Foundations of Quantum Theory
Brief: It presents the evolution of the quantum observables from their classical counterpart and distinctly analyses the meaning of observable in context to both the classical and the quantum realms, emphasising 1925's W. Heisenberg realization.

2 An Open-ended Story on Quantization. OSF Preprints. 2023. DOI:10.31219/osf.io/92bvp.
Keywords: Foundations of Quantum Theory
Brief: To provide a write-up, impressing on the evolutionary track of quantization of physical quantities such as Energy, momentum, and angular momentum. Additionally, it addresses the question of which fundamental quantity is responsible for quantizing other derived quantities.

1 The Coronal Heating Problem OSF Preprints. 2023. DOI: shorturl.at/LE30p
Keywords: Solar Corona, Magnetic Reconnections
Brief: An overview of the Sun's atmospheric temperature inversion and de- tailing the recent developments in resolving the coronal heating problem.

Internships

3 प्रमात्रिक गतिकी के चित्र [*The pictures of quantum dynamics* (Hindi)]
Guide: Dr. Vikash K Ojha, SVNIT, Surat, [May-Jul, 2022]–URL: shorturl.at/lvEIE
Brief: Physics Education for Hindi Readers. The work explores 3 quantum dynamics methods and explains analytical mechanics, quantum mechanics, and philosophy. It aims to educate the public about quantum dynamics and its historical context.

2 The front-end development of a dummy e-commerce website
Guide: Dr. Santosh Prabhakar, Internship Studio, Pune, [Jun-Jul, 2020]
Brief: Learning included using HTML5, CSS3 frameworks and training for the web.

1 Project, Building the student information management system using C language
Guide: Dr. Balu Parne, SVNIT, Surat, [Feb-May, 2020]
Brief: Learning included using C languages and its application in file handling.

Achievements

09/16-present:	NTS Scholarship of merit, Government of India, New Delhi.
03/24	Qualified Graduate Aptitude Test in Engineering (GATE)-2024
04/23	Qualified Graduate Aptitude Test in Engineering (GATE)-2023
02/20	Winner (Physics), InQuest 4.0 Researchathon, SCOSH [Student Chapter], SVNIT

Courses

University (Physics):	Quantum Field Theory I, Many-Body Physics and Relativistic Quantum Mechanics, Particle Physics, Nuclear Physics, Special Relativity, Quantum Mechanics I & II, Computational Physics(Python, Monte-Carlo), Computational Methods (MATLAB/Octave), Atomic and Molecular Physics, Electrodynamics, Statistical Mechanics, Astrophysics, Density Functional Theory, and Classical Mechanics.
University (Mathematics):	Singular perturbation theory, Special Functions, Linear Algebra, Tensors, Vector Calculus, Integral transforms, Real & Complex analysis, Computational Methods, Infinite Series, Solutions to ODEs and PDEs, Numerical methods, Interpolation, Partial Differentiation, solutions to linear equations systems etc.

Audit-courses:

- Algebraic Topology in Physics, Physics Latam, Prof. Pavel Putrov (ICTP) [05-06/24]
- A Short Course in Math History, Prof. A. Viramani (CMI) [08-09/21]
- Classical Electromagnetism, Prof. HC Verma (IITK) [08-12/20]
- Learning Physics through Simple Experiments, Prof. HC Verma (IITK) [03-06/20]
- Advanced course on Special Theory of Relativity, Prof. HC Verma (IITK) [01-05/20]
- Basics of Quantum Mechanics, Prof. HC Verma (IITK) [08-11/19]
- The basics of Special Theory of Relativity, Prof. HC Verma (IITK) [01-03/19]

Skill Sets

Programme:	C, Python, MATLAB, Octave, Mathematica
Softwares:	LaTeX, MS Excel, and use of GitHub.
Web Design:	Basics of HTML5, CSS3
OS:	Windows, Ubuntu
Languages:	English (Intermediate), Hindi(Native)

Workshops & Seminars

07/21:	*Quantum Fields, Geometry & Representation Theory*, ICTS-TIFR, Bangalore.
12/20:	National workshop on *Data analysis using MS Excel*, BBD NITM, Lucknow.
08/20:	National workshop on MATLAB *Tools And Applications*, BBD NITM, Lucknow.

Books & Talks

02/07/23:	मनोविघ्न: (MaNoVighn) Short Poetry Collections, Kindle, ASIN: B0C9Y7X1PG
01/09/21:	कालसंघर्ष: (KālSanGharsha) Short Poetry Collections, Kindle, ASIN: B09FC1XRPK
13/11/21:	*Unification in Mathematics*, IntERAct Seminar series, AMHD, SVNIT-Surat

Responsibilities & Positions

08/22-03/23	Student Coordinator: QUANTA Seminar Series, Department of Physics, SVNIT Surat
11/22-01/23:	Design-Infra Coordination Head & Member: *Manoj Memorial Night Cricket Tournament* (MMNCT) Annual cricket event, SVNIT Surat
10/21-07/22:	Joint Academic Affairs Secretary: *Academic Affairs Council* SVNIT, Surat
06/20-07/22	Co-convener (03/21) & Member *Society for Cultivation of Science and Humanities* (SCOSH) Student Chapter, SVNIT Surat
09/19-07/20	Student Volunteer: *Unnat Bharat Abhiyan* (UBA) SVNIT Surat

Other

I have always been an enthusiastic, independent seeker of truth. I will listen to mathematical beauty, ancient history, and philosophical ideas and stand up for social justice. My hobbies include Hindi Poetry, Graphic design, and writing articles.

Reference

Prof. Dr. Ajay Kumar Rai | email Id : akr@phy.svnit.ac.in| Professor, Physics, SVNIT
Dr. Vikash Kumar Ojha | email Id : vko@phy.svnit.ac.in| Assistant Professor, Physics, SVNIT

Declaration: *I hereby declare that the information given above is true to the best of my knowledge.*

Digitally Signed: Roshan Raj

Date: 30/05/2024.